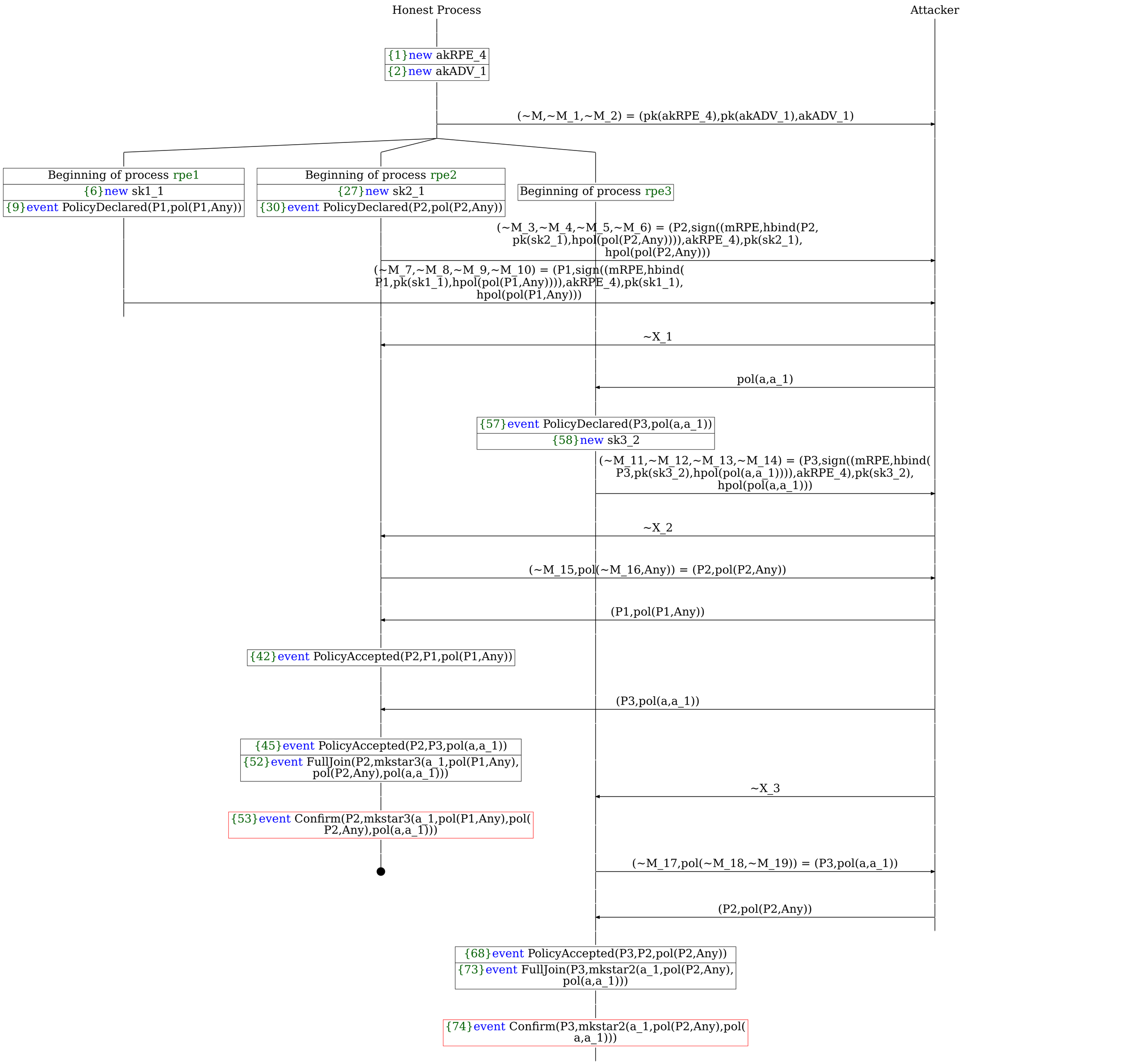
A trace has been found.

Abbreviations
~X_1 = (P1,~M_8,~M_9,hpol(pol(P1,Any))) = (P1,sign((mRPE,hbind(P1,pk(sk1_1),hpol(pol(P1,Any)))),akRPE_4),pk(sk1_1),hpol(pol(P1,Any)))
~X_2 = (P3,~M_12,~M_13,hpol(pol(a,a_1))) = (P3,sign((mRPE,hbind(P3,pk(sk3_2),hpol(pol(a,a_1)))),akRPE_4),pk(sk3_2),hpol(pol(a,a_1)))
~X_3 = (P2,~M_4,~M_5,hpol(pol(P2,Any))) = (P2,sign((mRPE,hbind(P2,pk(sk2_1),hpol(pol(P2,Any)))),akRPE_4),pk(sk2_1),hpol(pol(P2,Any)))